

Supplementary Information

Supplementary Table 1: Gini Indices for raw tree count and basal area, measured within DAs of the study cities. Five cities did not record tree diameter at breast height, so basal area could not be calculated.

City	Tree Count	Basal Area
Ajax	0.455	0.468
Burlington	0.303	0.487
Calgary	0.488	0.516
Edmonton	0.536	0.654
Fredericton	0.259	0.376
Guelph	0.350	0.472
Halifax	0.495	-
Kelowna	0.544	0.685
Kingston	0.492	0.493
Kitchener	0.409	0.414
Lethbridge	0.552	0.709
Longueuil	0.507	0.445
Maple Ridge	0.681	-
Mississauga	0.384	0.506
Moncton	0.476	0.547
Montreal	0.365	0.463
New Westminster	0.445	-
Niagara Falls	0.396	0.349
Ottawa	0.489	0.589
Peterborough	0.355	-
Quebec City	0.512	0.602
Regina	0.283	0.438
St. Catharines	0.309	0.404
Strathcona County	0.421	0.555
Toronto	0.402	0.493
Vancouver	0.348	0.443
Victoria	0.338	0.379
Waterloo	0.386	-
Welland	0.400	0.469
Whitby	0.339	0.392
Windsor	0.345	0.324
Winnipeg	0.379	0.420

Supplementary Table 2: Spatial lag (SLM) and spatial error (SEM) model results for street tree density.

Province	City	Model	Intercept	Population Density	Residential Instability	Ethno-Cultural Composition	Economic Dependency	Situational Vulnerability	S/L term	AIC	BIC
British Columbia	Maple Ridge	SLM						-151*	0.578***	1447.6	1468.8
	Vancouver	SEM	1.41e+03***	0.0172***	-121***				0.620***	15124.2	15163.3
Alberta	Calgary	SLM	241***	0.00391**		-43.0***	-34.9**		0.625***	24128	24170.7
	Edmonton	SLM	96.9***	0.0222***		-66.9***	-50.8***		0.754***	19350.5	19391.8
	Lethbridge	SLM							0.754***	1985.1	2008.3
Saskatchewan	Regina	SLM		0.104***	-85.8***	-164***		87.8***	0.624***	5518.9	5550.2
Manitoba	Winnipeg	SLM	214***	0.0280***	-147***	-40.1**	-46.9**	29.9*	0.706***	16277.1	16317.1
Ontario	Ajax	SLM		0.0786***	-359***				0.403***	2604.7	2629.7
	Burlington	SLM	443***	0.0290***	-199***	138*			0.407***	4134.5	4163.3
	Guelph	SLM		0.103***	-146***	-245***			0.644***	2799.8	2825.9
	Kitchener	SEM	768***	0.0757***	-288***	-166***			0.475***	4735.9	4765.9
	Mississauga	SLM	533***		-355***	131***			0.390***	14957.6	14996.4
	Ottawa	SLM	160***	0.0177***	-95.6***		-52.4**		0.731***	18650.8	18691.6
	St. Catharines	SEM	520***	0.107***	-207***	-223***			0.431***	3155.5	3182.7
	Toronto	SEM	1.66e+03***	0.0103***	-217***	-151***			0.595***	57431.5	57481.2
	Waterloo	SLM		0.163***	-255***	-133**	-125**		0.208*	2226.1	2250.3
	Welland	SLM	-219*	0.138***	-194***				0.752***	1196.1	1215.6
Quebec	Windsor	SLM	490***	0.0604***	-89.7***		-116***		0.342***	5433.9	5465.2
	Longueuil	SLM	389***						0.494***	6174.7	6206.8
	Montreal	SLM	407***	0.0173***	-76.3***	-106***	-38.9**		0.683***	37639.7	37686.1
Quebec	Quebec City	SLM		0.0186***			-82.7***		0.730***	12655.0	12692.9
New Brunswick	Fredericton	SLM		0.183***	-145***			-95.5**	0.403***	1120.1	1139.4
Nova Scotia	Halifax	SLM		0.0128**		-121***	-78.0***		0.777***	6556.6	6589.3

Supplementary Table 3: Spatial lag (SLM) and spatial error (SEM) model results for street tree median DBH.

Province	City	Model	Intercept	Population Density	Residential Instability	Ethno-Cultural Composition	Economic Dependency	Situational Vulnerability	S/L term	AIC	BIC
British Columbia	Kelowna	SLM	18.6***						0.363***	1332.0	1355.7
	Vancouver	SLM	12.3***		-0.0629*	-1.17***			0.596***	7234.4	7273.7
	Victoria	SLM	17.9***						0.432***	1016.9	1040.5
Alberta	Calgary	SLM	1.03***	9.10e-05*	-0.360*	-1.89***	0.951***	1.22***	0.578***	10915.6	10958.8
	Edmonton	SLM	8.74***			-2.96***		1.52***	0.687**	9786.2	9827.4
	Lethbridge	SLM	12.2**			-6.43*		4.53*	0.579***	1103.6	1126.9
	Strathcona	SLM	10.7*		-4.17*		3.47*	7.97***	0.690***	718.2	738.5
Saskatchewan	Regina	SLM	5.25***		-0.915*	-1.17*		1.72***	0.170***	2532.7	2564.0
Manitoba	Winnipeg	SLM	9.22***		-0.738*	-0.949**		0.989***	0.707***	7920.8	7960.8
Ontario	Ajax	SLM	8.58***		-1.97*	4.11***			0.662***	1138.8	1163.8
	Burlington	SLM	7.21***			-3.65***			0.699***	1807.7	1836.6
	Guelph	SLM	12.4***		-1.80*	-4.84***		4.84***	0.675***	1377.2	1403.2
	Kitchener	SLM	15.2***			-2.36**	3.90***		0.427***	2317.9	2347.9
	Mississauga	SLM	1.08***	-7.26e-05***	-0.654*	-1.40***	1.03**		0.592***	6014.3	6053.1
	Niagara Falls	SLM	11.3**			-6.59**	3.37*		0.527***	1042.2	1065.5
	Ottawa	SLM	8.55***			-2.03***	0.893***	1.56***	0.642***	8670.0	8710.8
	St. Catharines	SLM	10.3***		-1.95*	-4.94**	1.83*		0.617***	1606.7	1633.9
	Toronto	SLM	9.03***		-0.851***	-0.796***	0.327*	-0.587***	0.570***	24820	24869.4
	Whitby	SLM	10.4***			-4.37***	2.19**		0.568***	1281.4	1307.2
	Windsor	SLM	11.8***			-1.56**		2.33***	0.667***	2624.2	2655.5
Quebec	Longueuil	SLM	15.2***	5.14e-04*	-1.67**	-3.86***		3.21**	0.348***	2984.4	3016.5
	Montreal	SLM	10.6***	9.95e-05**	-1.71***	-0.738*	0.844**	1.52***	0.657***	19034.4	19080.8
	Quebec City	SEM	16.9***	6.67e-04***	-2.27***	-3.57***		2.65**	0.618***	6485.7	6523.7
New Brunswick	Fredericton	SLM	9.50**						0.479***	604.8	624.1
	Moncton	SLM		2.40e-03**					0.571***	797.6	818.6

Supplementary Table 4: Spatial lag (SLM) and spatial error (SEM) model results for street tree basal area (m³) per square kilometer.

Province	City	Model	Intercept	Population Density	Residential Instability	Ethno-Cultural Composition	Economic Dependency	Situational Vulnerability	S/L term	AIC	BIC
British Columbia	Vancouver	SLM	64.8***		-17.8***	-8.51*	-7.52*		0.705***	11888.6	11928.0
Alberta	Calgary	SLM	17.0***			-8.00***			0.721***	16491.5	16534.7
	Edmonton	SLM	7.18*	2.92e-03***		-13.4***	-7.19***	5.57*	0.842***	14506.3	14547.6
	Lethbridge	SLM							0.782***	1560.8	1584.0
	Strathcona	SLM	37.4*		-14.4*	30.3*	13.6**	31.5***	0.568***	959.9	980.2
Saskatchewan	Regina	SLM		0.0123***	-14.0**	-24.1***		17.6***	0.797***	4351.4	4382.8
Manitoba	Winnipeg	SLM	11.4**	0.003079***	-15.2***	-10.1***	-6.53**	8.31***	0.832***	12096.8	12136.8
Ontario	Ajax	SLM	24.0**	0.00159*		-14.9***			0.619***	1679.0	1704.0
	Burlington	SLM	18.0*		-11.1*	-26.4***			0.712***	2910.7	2939.6
	Guelph	SLM		0.00696***	-11.4*	-35.4***			0.734***	2059.3	2085.4
	Kitchener	SEM	70.6***	0.00777***	-29.4***	-22.4***			0.710***	3454.7	3484.7
	Mississauga	SLM	50.6***		-10.3***	-13.4**	4.18*		0.531***	9675.7	9714.5
	Ottawa	SLM	15.3***	0.00125***	-10.3***	-15.3***		6.21*	0.826***	13209.7	13250.5
	St. Catharines	SLM		0.0102***	-15.9*	-58.8***			0.580***	2476.9	2504.1
	Toronto	SLM	71.2***	0.000276**	-18.6***	-19.1***	-2.68*	-3.54*	0.660***	41447.1	41496.9
	Windsor	SLM	47.2***	0.00589***	-13.2**	-14.0***	-17.9***	11.6*	0.639***	4118.9	4150.2
Quebec	Longueuil	SLM	25.5***			-10.6*			0.569***	4251.1	4283.2
	Montreal	SEM	175***	0.00229***	-28.9***	-21.7***		11.0**	0.747***	28605.7	28652.1
	Quebec City	SLM		0.00314***	-4.81*	-10.8*	-6.65***		0.799***	9399.5	9437.5
New Brunswick	Moncton	SLM		0.0112***		-19.5*			0.697***	1066.5	1087.5

Supplementary Table 5: Ordinary least squares (OLS), spatial lag (SLM), and spatial error (SEM) model results for street tree species diversity.

Province	City	Model	Intercept	Population Density	Residential Instability	Ethno-Cultural Composition	Economic Dependency	Situational Vulnerability	S/L term	AIC	BIC
British Columbia	Kelowna	SLM	1.45***	-6.16e-05**			-0.107***		0.435***	273.8	299.8
	Maple Ridge	SLM	1.25***	-1.61e-04**	0.435***	0.667***		-0.531**	0.329**	228.4	249.6
	New Westminster	SLM	2.09***	-4.43e-05***			-0.175*		0.302*	154.31	174.3
	Vancouver	SLM	1.25***	-1.76e-05***				-0.127***	0.515***	1024.9	1064.3
Alberta	Calgary	SLM	1.13***	-2.50e-05***		-0.0315*		-0.130***	0.484***	1933.3	1976.5
	Edmonton	SLM	0.85***	-3.39e-05***				-0.0944***	0.393***	2386.1	2427.3
Manitoba	Winnipeg	SEM	1.91***	-3.19e-05***	-0.0659***			-0.0905***	0.631***	1474.2	1514.2
Ontario	Ajax	SLM	1.30***	-4.98e-05***		0.154*			0.371***	257.7	282.7
	Burlington	SLM	1.34***	-3.29e-05***					0.572***	211.9	240.8
	Guelph	SLM	1.45***	-6.16e-05**			-0.107***		0.435***	273.8	299.8
	Kingston	SEM	3.07***	-1.06e-04***				-0.124*	0.763***	221.9	246.8
	Kitchener	SLM	1.67***	-6.85e-05***			-0.139***	-0.170**	0.311***	480.2	510.3
	Mississauga	SLM	1.42***	-9.52e-06***	-0.0653***	-0.0912***	-0.0450*		0.460***	878.2	917.0
	Niagara Falls	OLS	2.28***	-1.10e-04**					-	168.0	188.3
	Ottawa	SEM	2.63***	-4.22e-05***	-0.0795***		-0.0393*	-0.183***	0.514***	1553.7	0.388
	St. Catharines	SLM	1.08***	-7.95e-05***	0.127***	-0.124*			0.590***	187.2	214.4
	Toronto	SLM	1.83***	-1.74e-05***	-0.0653***	-0.0987***	-0.0204*	-0.0428***	0.468***	3840.4	3890.1
Quebec	Windsor	SEM	2.86***	-3.13e-05***	-0.104***				0.629***	120.4	51.7
	Longueuil	SEM	2.91***	-6.37e-05***					0.448***	434.75	466.8
	Montreal	SEM	2.54***	-2.08e-04***	-0.0388**	-0.0445**	-0.0440***		0.567***	2818.9	2865.3
New Brunswick	Quebec City	SEM	2.63***	-6.93e-04***	-0.0481*	0.240***	-0.0491**	-0.149***	0.485***	1217.5	1255.4
	Fredericton	SEM	2.40***	-9.76e-05*				-0.234***	0.591***	93.3	112.5
	Moncton	SLM	1.12***	-1.08e-04**				-0.2267**	0.595***	116.5	137.5

Supplementary References 1: Tree Inventories Included in the Study

Canadian tree inventories included in this study. File types are either shapefiles (.shp) or comma separated values (.csv). The inventories are categorized by province, ranging from east to west. Because the inventories are continuously updated, the year of publishing is reported instead of the most recent revision. Note that as maintenance is performed on the datasets, some URLs may be temporarily unavailable.

British Columbia

- Kelowna. (n.d.). Tree inventory.shp. City of Kelowna. <https://opendata.kelowna.ca/datasets/tree-inventory/explore>
- Maple Ridge. (2018). Street trees.shp. City of Maple Ridge. Retrieved from <https://opengov.mapleridge.ca/datasets/mapleridge::street-tree/about>
- New Westminster. (2022). Tree inventory.shp. New Westminster. Retrieved from <https://opendata.newwestcity.ca/datasets/newwestcity::tree-inventory>
- Vancouver. (2024). Street trees.csv. City of Vancouver Open Data Portal. Retrieved from https://opendata.vancouver.ca/explore/dataset/public-trees/information/?disjunctive.neighbourhood_name&disjunctive.on_street&disjunctive.species_name&disjunctive.common_name
- Victoria. (2019). Tree species.shp. City of Victoria Open Data Portal. Retrieved from <https://opendata.victoria.ca/datasets/36e90771770542baaa89afddce69195a/explore?location=48.419018%2C-123.350267%2C14.98>

Alberta

- Calgary. (2018). Public trees.shp. City of Calgary's Open Data Portal. Retrieved from https://data.calgary.ca/Environment/Public-Trees/tfs4-3wwa/about_data
- Edmonton. (2012). Trees.shp. City of Edmonton's Open Data Portal. Retrieved from https://data.edmonton.ca/Environmental-Services/Trees/eecg-fc54/about_data
- Lethbridge. (2017). Trees.shp. City of Lethbridge Open Data Catalogue. Retrieved from https://opendata.lethbridge.ca/datasets/82841132047d47659508f60c52f6346a_0/about
- Strathcona County. (2024). Trees.shp. Strathcona County Open Data. Retrieved from https://opendata-strathconacounty.hub.arcgis.com/datasets/8a68e1b6c525481bbd2b524616739ee1_0/explore?location=53.519503%2C-113.368132%2C10.96

Saskatchewan

- Regina. (2019). City of Regina tree inventory geospatial shapefile.shp. City of Regina. Retrieved from <https://open.regina.ca/dataset/city-of-regina-tree-inventory/resource/e354a39b-db42-4674-9f56-bf2d87aa841e>

Manitoba

- Winnipeg. (2017). Tree inventory.shp. City of Winnipeg Open Data. Retrieved from https://data.winnipeg.ca/Parks/Tree-Inventory/hfwk-jp4h/about_data

Ontario

- Ajax. (2017). Town trees.shp. Ajax Open Data. Retrieved from https://opendata.ajax.ca/datasets/c5d5ff5351a54fdb8d3934abbb5cda9a_8/explore?location=43.866306%2C-79.023600%2C13.09
- Burlington. (2017). City owned trees.shp. Navigate Burlington. Retrieved from <https://navburl-burlington.opendata.arcgis.com/datasets/41412ef3188942c7abbe14035004273d/explore>
- Guelph. (2017). Tree inventory.shp. Guelph Open Data. Retrieved from <http://data.open.guelph.ca/datafiles/gis/TreeInventory.zip>
- Kingston. (2023). City-owned trees.shp. Open Data Kingston. Retrieved from https://opendatakingston.cityofkingston.ca/explore/dataset/trees-municipal/information/?disjunctive.owner&disjunctive.common_name&disjunctive.scientific_name
- Kitchener. (2017). Tree inventory.shp. Kitchener GeoHub. Retrieved from <https://open-kitchenergis.opendata.arcgis.com/datasets/e42a9601b87a4417b3d3d15a0f596f64/explore>
- Mississauga. (2023). 2023 city owned tree inventory.shp. Mississauga Open Data. Retrieved from <https://data.mississauga.ca/datasets/mississauga::2023-city-owned-tree-inventory/explore>
- Niagara Falls. (2023). Niagara Falls trees inventory.shp. City of Niagara Falls Open Data. Retrieved from https://open.niagarafalls.ca/datasets/24fa04710ac9481f834aa180ebcabb41_0/explore
- Ottawa. (2018). Tree inventory.shp. Open Ottawa. Retrieved from <https://open.ottawa.ca/datasets/tree-inventory/explore>
- Peterborough. (2022). Tree inventory.shp. City of Peterborough GIS Open Data. Retrieved from <https://data-ptbo.opendata.arcgis.com/datasets/tree-inventory/explore>
- St. Catharines. (2024). St. Catharines trees.csv. Niagara Open Data. Retrieved from <https://niagaraopendata.ca/dataset/st-catharines-trees>
- Toronto. (2024). Street tree data.shp. City of Toronto Open Data Portal. Retrieved from <https://open.toronto.ca/dataset/street-tree-data/>
- Waterloo. (2018). Street tree inventory.shp. Region of Waterloo – Open Data. Retrieved from <https://rowopendata-rmw.opendata.arcgis.com/datasets/City-of-Waterloo::street-tree-inventory/explore?location=43.477383%2C-80.540470%2C13.00>
- Welland. (2021). Welland trees.shp. City of Welland Open Data Portal. Retrieved from <https://open.welland.ca/datasets/afcb6b5308de40b1830a3ebb2369a83b/explore>
- Whitby. (2020). Trees.shp. Whitby GeoHub. Retrieved from https://geohub-whitby.hub.arcgis.com/datasets/5be6efcf4f264639874bf07009f155d2_0/explore?location=44.120901%2C-79.338592%2C8.62
- Windsor. (2022). Trees in right of way.csv. City of Windsor Open Data Catalogue. Retrieved from <https://opendata.citywindsor.ca/details/238>

Quebec

- Longueuil. (2020). Arbres.shp. Ville de Longueuil GéoHub. Retrieved from <https://geohub.longueuil.quebec/datasets/arbres/explore>
- Montréal. (2020). Arbres publics sur le territoire de la Ville.csv. Montréal Données ouvertes. Retrieved from <https://donnees.montreal.ca/dataset/arbres>
- Ville de Québec. (2012). Arbres répertoriés.shp. Partenariat Données Québec. Retrived from https://www.donneesquebec.ca/recherche/dataset/vque_26

New Brunswick

- Fredericton. (2022). Tree inventory for public use.shp. Fredericton Open Data. Retrieved from <https://data-fredericton.opendata.arcgis.com/datasets/25c9ca6a32a740428adb225a11783c38/explore>
- Moncton. (2018). Trees.shp. City of Moncton Open Data. Retrieved from <https://open.moncton.ca/datasets/b198ccbbd170480dadb073246f159138/explore>

Nova Scotia

- Halifax. (2020). Public trees.shp. Halifax Regional Municipality Open Data. Retrieved from https://data-hrm.hub.arcgis.com/datasets/33a4e9b6c7e9439abcd2b20ac50c5a4d_0/explore

Supplementary References 2: Road Network Shapefiles

Twelve tree inventories included park trees but did not include a column identifying trees as boulevard/street versus park trees. To remove non-street trees, buffers from road networks were used. One of two buffer distances from road centerlines were used. If the road network included a column identifying the number of lanes, the equation

$$12 \text{ meters} + 4 \text{ meters} \cdot \text{lane count}$$

was used. If the road network shapefile did not include a column for lane count, then a buffer of 20 meters was used.

File types are either shapefiles (.shp) or comma separated values (.csv). Because the road networks are continuously updated, the year of publishing is reported instead of the most recent revision. Note that as maintenance is performed on the datasets, some URLs may be temporarily unavailable.

Cities with Lane Counts

- Burlington. (2017). Road segments.shp. Navigate Burlington. Retrieved from https://navburl-burlington.opendata.arcgis.com/datasets/ffcc3410ffb047949afb0422f5596188_0/explore
- Fredericton. (2024). Street center lines.shp. Fredericton Open Data. Retrieved from https://data-fredericton.opendata.arcgis.com/datasets/d3f6933108cc4c439314571e2078ac9d_15/explore?location=45.941580%2C-66.648738%2C12.20
- Halifax. (2014). Street centrelines.shp. Halifax Regional Municipality Open Data. Retrieved from https://data-hrm.hub.arcgis.com/datasets/560fec412dd044b08ae52a8575a215d4_0/explore?location=44.811541%2C-63.229831%2C10.77
- Kingston. (2023). Road centerline.shp. Open Data Kingston. Retrieved from https://opendatakingston.cityofkingston.ca/datasets/d6bac08f76db4d319f65e0cd3ad7c9db_18/explore

Cities without Lane Counts

- Kelowna. (2017). Road.shp. City of Kelowna GIS. Retrieved from https://opendata.kelowna.ca/datasets/c5a9895698944b85ad0b9a7e4294573b_5/explore
- New Westminster. (2020). Street centerline.shp. New Westminster Open Data Portal. Retrieved from https://opendata.newwestcity.ca/datasets/d3a0a6b32a2e4bf4a72c0badf2fbf04d_0/explore?location=49.206909%2C-122.918192%2C13.81
- Strathcona County. (2024). Street network.shp. Strathcona County Open Data. Retrieved from https://opendata-strathconacounty.hub.arcgis.com/datasets/e954674d8096432b922b2a0364859055_0/explore
- Regina. (2017). Road centerline.shp. City of Regina. Retrieved from <https://open.regina.ca/dataset/road-centerline>

- Ajax. (2001). Ontario road network (ORN) road net element.shp. Ontario GeoHub. Retrieved from https://geohub.lio.gov.on.ca/datasets/2fd52bccdb77479da0133c86545503f8_0/explore?location=43.853444%2C-79.036734%2C16.81
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